

Health and Economic Returns to Dietary Sodium Reduction

Technical review of the source-split CVD model and preliminary results for prioritized countries

Resolve to Save Lives (RTSL) and University of Washington

July 2026

Recap and today's plan

Recap: where we left off (May 28 call)

From the previous presentation and discussion:

- **Modeling framework** — multi-state CVD life-table (Well → Sick → Dead), simulated annually for ages 20–95, by age × sex × country, for four causes: IHD, ischemic stroke, hemorrhagic stroke, hypertensive heart disease.
- **Outcome metrics** — deaths and cases delayed, DALYs averted, plus a full benefit–cost analysis (NPV, BCR).
- **Causal pathway** — sodium → systolic BP (Filippini dose–response) → CVD incidence (Ettehad per-10-mmHg relative risks) → life-table transitions (Filippini et al. 2021; Ettehad et al. 2016).
- **Interventions** — six policy levers scored in the Sodium Lives Saved Calculator (public procurement, media, low-sodium salt substitutes, front-of-pack labelling, salt targets, fiscal).
- **Deliverables** — peer-reviewed paper, RTSL report, and interactive Shiny tool.

Today we present the **current build** of the model and **preliminary results** for the prioritized countries.

Recap: what the May 28 discussion decided

Scope & data

- Fixed to **10 priority countries** (down from the broader ~29 / PAHO list).
- Sodium sources: **combined approach** — use country data where available, else the Bhat/Marklund GDP regression (Bhat et al. 2020). Drivers extend beyond GDP (condiments/bouillon in Viet Nam, Nigeria).
- **Do not map interventions to SHAKE** — no longer how the revised SHAKE frames them.

Policies & framing

- **Fiscal policy**: feasibility & generalizability doubts; RTSL not interested in salt taxes.
- **LSSS**: focus scenarios 2/4/5; include potassium.
- **Media**: revised to a UK basis; may be dropped.
- Goal is not only to *maximize* lives saved (e.g. prevention framing for Ethiopia).
- Primary deliverable: **BP reinvestment proposal** (memo, tables + figures), **Sept 2026**.

Today's agenda

1. **Model inputs** — updated sodium source shares by country (now incl. Nigeria/Cameroon) and the applied salt-effect η for each intervention and the full package.
2. **Methods** — the source-split logic (sources $\rightarrow \eta$), the state-transition engine, and the current low-sodium salt substitute (LSS) model.
3. **Preliminary results** — baseline burden, projected trajectories by scenario, and deaths delayed across the 10 prioritized countries.
4. **Limitations.**
5. **Discussion** — pending decisions (fiscal in/out, LSS scenarios, source-share strategy, harmonizing observation periods) and the timeline to the September reinvestment deliverable.

Background

Project objective and policy question

- Estimate the **national-scale health and economic impact** of dietary sodium-reduction policies using a population CVD model.
- Quantify **deaths and cases delayed** (IHD, ischemic stroke, hemorrhagic stroke, hypertensive heart disease) and **DALYs averted** over 2026–2050, with a full **benefit–cost analysis**.
- Focus on **RTSL partner countries** and on policies governments can realistically implement.

The question the model answers

How many cardiovascular deaths are *delayed* by each sodium-reduction policy — and by a coordinated **package** — when each policy's documented effect is applied **only to the sodium sources it can actually reach**?

Causal pathway and scope

Exposure → disease chain (unchanged):

Sodium → SBP → CVD incidence → Well → Sick → Dead

- **SBP response** to sodium from the Filippini dose–response meta-analysis (Filippini et al. 2021).
- **Incidence response** to SBP from the Ettehad per-10-mmHg relative risks (Ettehad et al. 2016), applied by BP bin and cause.
- **Four causes** modelled jointly (IHD, ischemic stroke, hemorrhagic stroke, HHD); **case fatality is held at baseline**, so the model shifts deaths later — we report **deaths delayed**, not “averted”.
- **Scope:** 10 prioritized countries; annual projection over **2017–2050**; scale-up 2026–2030.

Model inputs

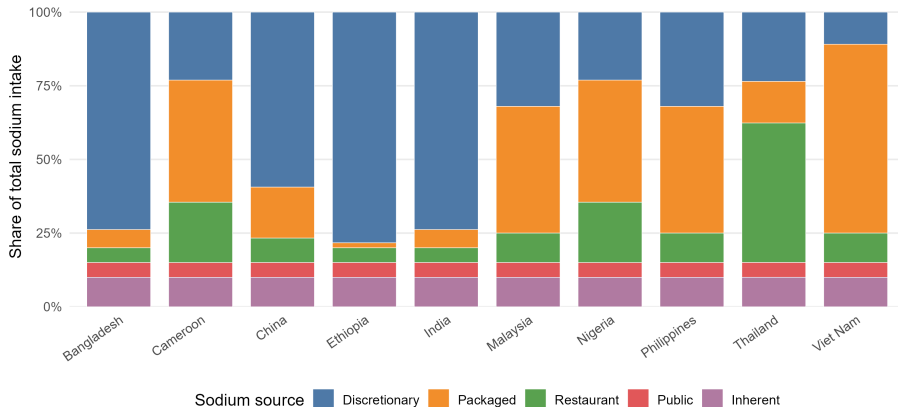
Model inputs at a glance

Input	Role
Baseline transition rates (GBD)	Age $\times$ sex $\times$ year $\times$ cause incidence (IR), case fatality (CF), background & COVID mortality, population (N_x).
BP distribution	Mean/SD systolic BP by age $\times$ sex, raised-BP fraction, diabetes prevalence.
Baseline sodium intake	2024 mean intake, split into five sources.
Sodium source shares	Five-source split of baseline sodium per country (this section).
GBD relative risks	Log-linear RR per 10 mmHg SBP; sets bin-specific baseline incidence.
Ettehad relative risks	RR per 10 mmHg SBP reduction by BP bin $\times$ cause.
Scale-up schedule	Linear ramp 2026 $\rightarrow$ 2030, then held.

The source-split logic is the only change from the earlier single-pool build; everything downstream of η is unchanged.

Sodium source shares by country

Baseline sodium is partitioned into **five mutually exclusive sources** (shares sum to 1). **inherent sodium is never targeted**; each policy acts only on the source(s) it can reach.



Country data where available; otherwise predicted from GDP per capita (Bhat et al. 2020). **Nigeria/Cameroon** use provisional

Applied salt-effect (η) by intervention

Each intervention has a documented **relative reduction** applied to its **target source**. Values match `code/07_run_interventions.R` exactly.

Intervention	Source	Applied reduction	Note
Public food procurement & service	public	20.0%	
Behaviour change / media campaigns	discretionary	1.0%	
Behaviour change / media campaigns	packaged	1.0%	
Front-of-pack labelling (FOPL)	packaged	13.8%	
Salt targets (reformulation)	packaged	20.0%	
Fiscal policy (salt tax)	packaged	0.0%	
Low-sodium salt substitutes (LSS)	discretionary	15.0%	SSaSS-derived; hypertensive reach; in package

Media acts on discretionary **and** packaged (1.0% each). FOPL, salt targets and fiscal all act on **packaged** and compose multiplicatively when combined: e.g. FOPL (13.8%) + salt targets (20%) $\Rightarrow 1 - (1 - 0.138)(1 - 0.20) = \mathbf{31.0\%}$. **Fiscal is currently 0% — excluded pending the team's decision.**

Illustrative η by intervention and full package

$\text{salteff}(\eta)$ is the **fraction of total intake** each scenario reduces at full implementation — **country-specific**, because it depends on local source shares:

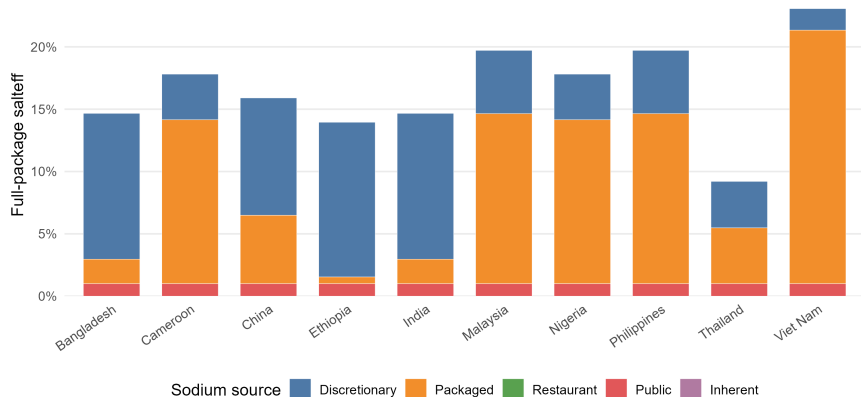
$$\eta = \sum_s \text{share}_s \cdot \left[1 - \prod_i (1 - \text{eff}_{i,s}) \right].$$

Scenario	Viet Nam	Bangladesh	China	Thailand	default
Public food procurement & service	0.0100	0.0100	0.0100	0.0100	0.0100
Behaviour change / media campaigns	0.0075	0.0080	0.0077	0.0038	0.0071
Front-of-pack labelling (FOPL)	0.0885	0.0086	0.0239	0.0195	0.0384
Salt targets (reformulation)	0.1282	0.0124	0.0346	0.0282	0.0557
Fiscal policy (salt tax)	0.0000	0.0000	0.0000	0.0000	0.0000
Low-sodium salt substitutes (LSS)	0.0164	0.1107	0.0891	0.0353	0.0645
Full package	0.2307	0.1466	0.1590	0.0921	0.1665

η is largest where the target source dominates local intake — packaged policies in packaged-heavy Viet Nam; LSS in discretionary-heavy Bangladesh/China. The full-package row composes all active policies.

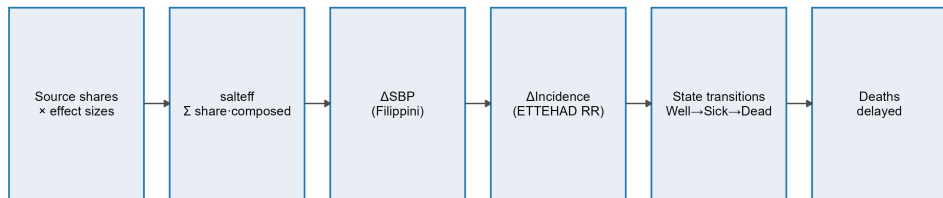
Full-package η : composition by country

Bar height = each country's full-package salt_{eff} ; segments show the per-source contributions ($\text{share}_s \times \text{composed}_s$).



Methods

The source-split approach: from sources to η



- Split baseline sodium into five sources; apply each policy's reduction **only** to the source(s) it reaches.
- Effects on the **same** source compose multiplicatively — composed_s = $1 - \prod_i (1 - \text{eff}_{i,s})$ — so overlapping packaged policies never double-count or exceed 100%.
- Share-weight and sum to a single total-intake fraction η (salteff), which feeds the unchanged epidemiological pathway.

Worked check (asserted in code): public procurement in Viet Nam touches only the public source (share 0.05, effect 20%)

$$\Rightarrow \eta = 0.20 \times 0.05 = \mathbf{0.0100}.$$

From η to health impact

Given a country's η , sodium falls and — through blood pressure — so does CVD **incidence** (case fatality is unchanged):

1. **Target reduction:** $\text{salt_target} = \text{salt} \cdot \eta$, subject to a **2 g/day minimum-intake floor**.
2. **Linear scale-up** from 2026 to 2030, then held.
3. **Sodium** $\rightarrow$ **SBP (Filippini)** (Filippini et al. 2021): $\Delta SBP = (2.8 \cdot \text{raisedBP} + 1.0 \cdot (1 - \text{raisedBP})) \cdot \text{salt_reduction}$.
4. **Baseline bin-specific incidence** scaled by the GBD log-linear RR per 10 mmHg, normalised to recover the observed IR .
5. **SBP** $\rightarrow$ **incidence (Ettehad)** (Ettehad et al. 2016): bin-specific RR per 10 mmHg drives the proportional incidence change; averaged over BP bins.
6. **Case fatality unchanged** $\Rightarrow$ **deaths delayed**, not averted.

The Lives Saved model: state-transition engine

Each country/sex/cause/scenario tracks **Well / Sick / Dead**, advanced one year at a time (ages 20–95, 2017–2050). With last year's cohort values:

$$\text{newcases}_t = \text{well}_{t-1} \cdot IR$$

$$\text{sick}_t = \text{sick}_{t-1} (1 - (CF + \text{BG.mx} + \text{covid.mx})) + \text{well}_{t-1} \cdot IR$$

$$\text{dead}_t = \text{sick}_{t-1} \cdot CF$$

$$\text{pop}_t = \text{pop}_{t-1} - \text{all.mx}_{t-1}, \quad \text{well}_t = \text{pop}_t - \text{all.mx}_t - \text{sick}_t$$

- Seeded (2017) from GBD prevalence and mortality; every quantity floored at zero.
- The **only** intervention lever is IR : sodium reduction lowers incidence via steps 1–6; CF , background and COVID mortality are untouched.
- **Aggregation**: the four causes share one population, so pop/all.mx are identical across cause rows (denominator taken from a single cause); $\text{sick/well/dead/newcases}$ are cause-specific and summed over causes.

Low-sodium salt substitute (LSS) model — current version

LSS is no longer a placeholder: it is derived from the **Salt Substitute and Stroke Study** (SSaSS) (Neal et al. 2021).

- **Effect size:** a 75% NaCl / 25% KCl substitute delivered an ITT net **-350 mg/day** sodium reduction against a **4.3 g/day** baseline $\Rightarrow$ **15% of discretionary sodium** (sensitivity 0.12–0.18). Applied to the discretionary source.

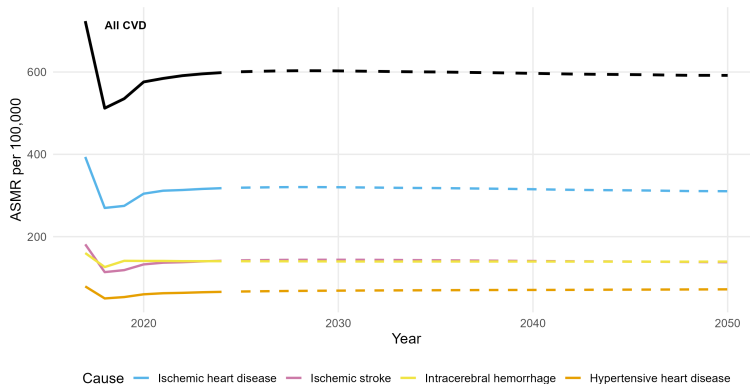
Two deliberately conservative choices

- **Hypertensive-only reach.** As its own scenario, the reduction applies only to raised-BP bins ($\text{SBP} \geq 140$); in the *full package* it composes across all bins.
- **No potassium channel.** SSaSS also raised urinary potassium (+803 mg/day), which drove much of the observed -3.34 mmHg SBP effect. With no K^+ pathway, the model captures only the sodium-mediated benefit — a **lower bound**.

Per the May call: include potassium in future work; focus on discretionary scenarios 2/4/5 (Huang et al. 2026).

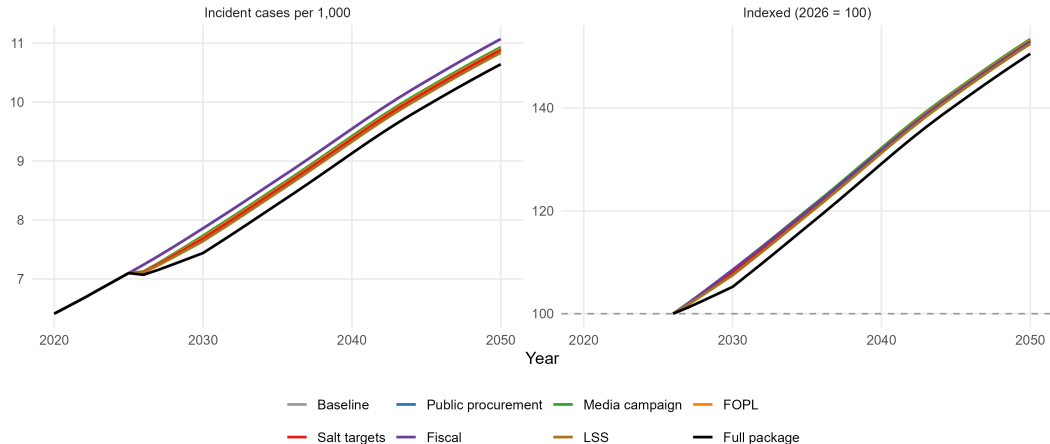
Baseline and projections

Baseline CVD burden, 2017–2050



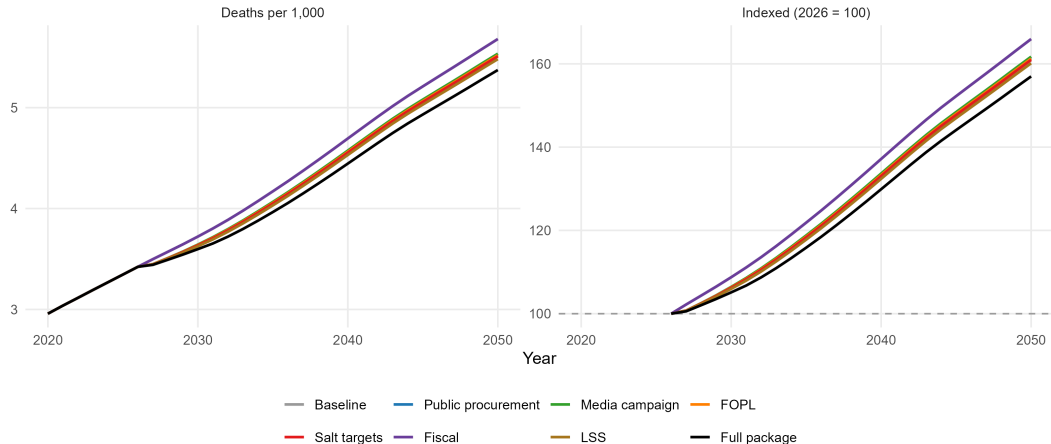
Age-standardised mortality (WHO standard, ages 20–95+) declines with forecasted background-mortality and case-fatality trends; absolute CVD deaths still rise with population growth and ageing. Solid = historical, dashed = projected.

Projected incidence by scenario



Incident CVD cases per 1,000 (left) and indexed to 2026 = 100 (right). Scenarios coincide until scale-up begins, then fan out; the **full package** falls fastest. **Fiscal overlaps baseline** because its applied η is currently 0%.

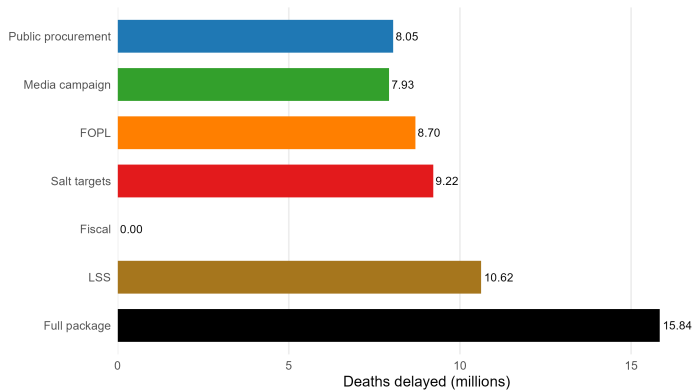
Projected deaths by scenario



CVD deaths per 1,000 (left) and indexed to 2026 = 100 (right). Because mortality responds to the **prior** year's incidence, the scenarios' divergence lags the incidence divergence by about a year.

Main results

Deaths delayed by scenario, 2026–2050



Headline

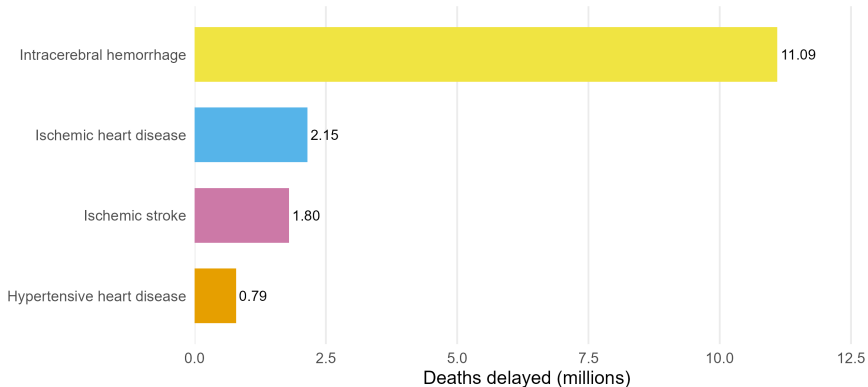
The ****full package**** delays ****15.84 million**** CVD deaths across 10 countries (4.6

Deaths delayed by scenario — detail

Scenario	Deaths delayed (millions)	Avg per year (thousands)	% of baseline
Public food procurement & service	8.051	322.0	2.32%
Behaviour change / media campaigns	7.931	317.3	2.28%
Front-of-pack labelling (FOPL)	8.696	347.8	2.50%
Salt targets (reformulation)	9.221	368.8	2.65%
Fiscal policy (salt tax)	0.000	0.0	0.00%
Low-sodium salt substitutes (LSS)	10.618	424.7	3.05%
Full package	15.835	633.4	4.55%

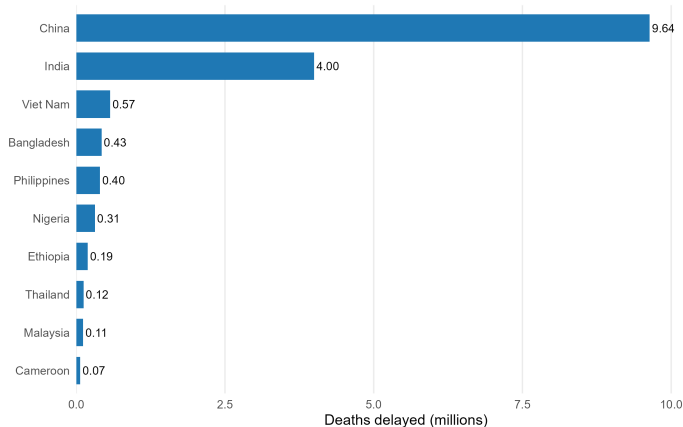
Among single policies, **LSS** delays the most (10.62 M) — it targets discretionary salt, which dominates intake in the most populous countries — followed by **salt targets** (9.22 M), **FOPL** (8.70 M), **public procurement** (8.05 M) and **media** (7.93 M). **Fiscal** delays 0.00 M (applied $\eta = 0\%$).

Full package: deaths delayed by cause



Ischemic heart disease and stroke dominate the delayed deaths, tracking their share of the baseline CVD burden and their SBP-mediated relative risks.

Full package: deaths delayed by country



Absolute counts are dominated by the most populous countries (China, India); per-capita impact depends on baseline burden and local source structure.

Limitations

Limitations and interpretation

- **Deaths delayed, not averted.** Case fatality is held at baseline, so the intervention lowers incidence and shifts mortality later; totals depend on the accounting window.
- **Point estimates only.** The per-source reductions are deterministic — no probabilistic sensitivity analysis yet.
- **LSS is a conservative lower bound** — sodium arm only, no potassium channel; hypertensive-only reach as an individual scenario.
- **Fiscal policy currently excluded** (applied $\eta = 0$)
- **Provisional source shares.** Nigeria/Cameroon values are working estimates; several others rest on single-study or older survey data.
- **Heterogeneous observation periods.** Documented effects span 1–20 year windows and are not yet harmonized onto a common footing.
- **Independent-cause structure.** The four causes share one population but are modelled independently.

Discussion and next steps

Pending decisions — for discussion

Open items

1. **Fiscal policy** — in or out? Currently $\eta = 0$
2. **LSS scenarios** — confirm discretionary scenarios 2/4/5; add a potassium channel to capture the full SSaSS effect?
3. **Observation-period harmonization** — effects measured over 1–20 yr. Apply as end-state at full scale-up, linearly over the observed window, or via a country-tier heuristic?
4. **Source-of-sodium strategy** — how hard to push for country-specific shares vs. the Bhat/Marklund GDP regression (Bhat et al. 2020)? Flag condiment/bouillon-heavy countries.
5. **Policy framing** — lives-saved maximization vs. a prevention/counterfactual framing (e.g. Ethiopia).
6. **Media / behavior change** — keep or drop?

Next steps and timeline

- **Lock the inputs** — finalize source shares (incl. Nigeria/Cameroon), confirm effect sizes, and resolve the fiscal decision.
- **Add the economic layer** — policy costing plus benefit–cost outputs (NPV, BCR) and DALYs.
- **Primary deliverable** — the **BP reinvestment proposal** memo (tables + figures), target **September 2026**.
- **Paper** is secondary; coordinate with Matti on the Bhat/Marklund inputs to avoid scooping the forthcoming paper.

Discussion

Where should we push hardest before the reinvestment deliverable — data (source shares), the fiscal/LSS decisions, or the economic layer?

Appendix

Appendix: per-intervention methods

The following slides document each intervention's **evidence base** and the **applied** η used in the model. Applied values are produced by `reports/report.RMD` and `mirror code/07_run_interventions.R`; study effect sizes are cited from source.

Also in this appendix: documented source-share values (Table M1) and the per-scenario η decomposition by country.

A. Public food procurement

Target source: public (institutional meals). **Applied η :** 20% of the public source.

- **Evidence:** Gase et al. (2011) — modeling study, LA County. Weighted **-233 mg/day** ($\approx 7\%$ of public-meal sodium).
- **Source adjustment:** for a country with share x of sodium from institutional meals, population impact $= 0.20 \times x$. Worked: Viet Nam $= 0.20 \times 0.05 = 0.0100$.
- **Caveats:** modeled (not observed); US-specific setting; small affected population share.

B. Behavior change / mass media

Target source: discretionary + packaged. **Applied η : 1.0% to each.**

- **Evidence:** Do et al. (2016) — COMBI Vietnam, **5%** fall in mean sodium after a 1-year multi-component campaign. Collins et al. (2014) — UK economic evaluation, **2%** [1–5%].
- The estimate was revised toward a UK basis; a conservative 1.0% is applied to each of discretionary and packaged.

Open question

RTSL funds **policy change, not mass media for behavior change** (Laura). Keep this lever in the model, or drop it?

C. Low-sodium salt substitutes (LSS)

Target source: discretionary salt. **Applied η : 15% of discretionary** (sensitivity 0.12–0.18).

- **Evidence:** Neal et al. (2021) (SSaSS) — -350 mg/day ITT sodium reduction vs. a 4.3 g/day baseline; observed -3.34 mmHg SBP (much of it potassium-driven, +803 mg/day).
- **Modelling:** sodium arm only (no K^+ channel) $\Rightarrow$ conservative lower bound; hypertensive-only reach as an individual scenario.
- **Scenarios** (Huang et al. 2026): focus on discretionary-only 2 (all), 4 (diagnosed HTN), 5 (treated HTN).

D. Front-of-pack labeling

Target source: packaged (estimate already from packaged-food purchases). **Applied η :** 13.8% of packaged.

- **Evidence:** Taillie et al. (2024) — Chile's mandatory warning-label law (observed). **-73.3 mg/capita/day (-13.8%)** in sodium purchases (2,844 households, 2013–2019).
- **Caveat:** purchases-to-intake translation is uncertain.
- **Alternative:** Mexico found $\approx -7\%$ in packaged sales (≈ -38 mg/capita/day).

E. Mandatory salt targets (reformulation)

Target source: packaged. **Applied η :** **20% of packaged** (conservative; 15% floor for sensitivity).

- **Policy:** South African mandatory maximum sodium limits (2013; targets implemented 2016 and 2019).
- **Evidence:** Strauss-Kruger et al. (2022) — African-PREDICT, **−15.2%** sodium; baseline/follow-up straddle the target dates, so likely an **underestimate**.
- **Alternatives:** (Gaziano et al. 2025) −8.4% (rural, spot urine); (Charlton et al. 2021) −18.4%; UK (He, Pombo-Rodrigues, and MacGregor 2013; Song et al. 2023); Mexico modeling (Muciño-Sandoval, Barrientos-Gutiérrez, et al. 2025).

F. Fiscal policy / taxation

Target source: packaged. **Applied η : 0% — currently excluded (see below).**

- **Evidence:** Saxena et al. (2025) — Philippines, modeled 20% tax on foods high in salt/sugar; **−100 mg/day (−10%)** at year 20.
- **Why excluded (for now):** political feasibility of food-category taxes; generalizability of a country-specific modeled estimate; RTSL not interested in salt taxes (May call).
- **Alternatives:** India (Roche et al. 2026); Australia (Davies et al. 2026; Cobiac et al. 2017); systematic review (Dodd et al. 2020) (1–11% reduction).

Appendix: sodium source shares — documented values

Country	Discretionary	Packaged	Restaurant	Public	Inherent
Viet Nam	0.109	0.641	0.100	0.050	0.100
Philippines	0.320	0.430	0.100	0.050	0.100
Bangladesh	0.738	0.062	0.050	0.050	0.100
China	0.594	0.173	0.083	0.050	0.100
Ethiopia	0.783	0.017	0.050	0.050	0.100
India	0.738	0.062	0.050	0.050	0.100
Malaysia	0.320	0.430	0.100	0.050	0.100
Thailand	0.235	0.141	0.473	0.050	0.100
Nigeria	0.231	0.415	0.205	0.050	0.100
Cameroon	0.231	0.415	0.205	0.050	0.100
default	0.430	0.279	0.142	0.050	0.100

Fractions of total intake, renormalised to 1.0. The default row (mean of the ten) applies to any country without its own row.

Appendix: η decomposition by scenario and country



Per-scenario η for every country, stacked by source contribution (note free y-axes). Fiscal is empty because its applied

η is currently 0%.

References

References

 Bhat, Saiuj et al. (2020). “A Systematic Review of the Sources of Dietary Salt Around the World”. In: *Advances in Nutrition* 11.3, pp. 677–686. DOI: 10.1093/advances/nmz134. URL: <https://www.sciencedirect.com/science/article/pii/S2161831322002927>.

 Charlton, Karen E. et al. (2021). “Leveraging Ongoing Research to Evaluate the Health Impacts of South Africa’s Salt Reduction Strategy: A Prospective Nested Cohort Within the WHO-SAGE Multicountry, Longitudinal Study”. In: *Current Developments in Nutrition*. URL: <https://www.sciencedirect.com/science/article/pii/S2211335521001595?via%3Dihub>.

 Cobiac, Linda J. et al. (2017). “Taxes and Subsidies for Improving Diet and Population Health in Australia: A Cost-Effectiveness Modelling Study”. In: *PLoS Medicine* 14.2, e1002232. DOI: 10.1371/journal.pmed.1002232. URL: <https://journals.plos.org/plosmedicine/article?id=10.1371/journal.pmed.1002232>.

 Collins, Marissa et al. (2014). “An Economic Evaluation of Salt Reduction Policies to Reduce Coronary Heart Disease in England: A Policy Modeling Study”. In: *Value in Health* 17.5, pp. 517–524. DOI: 10.1016/j.jval.2014.03.1722. URL: [https://www.valueinhealthjournal.com/article/S1098-3015\(14\)01828-2/fulltext](https://www.valueinhealthjournal.com/article/S1098-3015(14)01828-2/fulltext).

 Davies, Lauren et al. (2026). “Estimated Health and Economic Impact of an Excess Salt Tax in Australia: A Modelling Study”. In: *The Lancet Public Health*. DOI: 10.1016/S2468-2667(26)00043-5. URL: <https://www.thelancet.com/journals/lanpub/article/PIIS2468-2667%2826%2900043-5/fulltext>.

 Do, Hoang Thi Phuong et al. (2016). “Effectiveness of a Communication for Behavioral Impact (COMBI) Intervention to Reduce Salt Intake in a Vietnamese Province Based on Estimations From Spot Urine Samples”. In: *Journal of Clinical Hypertension* 18.11, pp. 1135–1142. DOI: 10.1111/jch.12884. URL: <https://onlinelibrary.wiley.com/doi/10.1111/jch.12884>.

 Dodd, Rebecca et al. (2020). “Effectiveness and Feasibility of Taxing Salt and Foods High in Sodium: A Systematic Review of the Evidence”. In: *Advances in Nutrition*. DOI: 10.1093/advances/nmaa067. URL: <https://www.sciencedirect.com/science/article/pii/S2161831322003520?via%3Dihub>.

 Ettehad, Dena et al. (2016). “Blood Pressure Lowering for Prevention of Cardiovascular Disease and Death: A Systematic Review and Meta-Analysis”. In: *The Lancet* 387.10022, pp. 957–967. DOI: 10.1016/S0140-6736(15)01225-8. URL: [https://www.thelancet.com/journals/lancet/article/PIIS0140-6736\(15\)01225-8/fulltext](https://www.thelancet.com/journals/lancet/article/PIIS0140-6736(15)01225-8/fulltext).

 Filippini, Tommaso et al. (Apr. 2021). “Blood Pressure Effects of Sodium Reduction: Dose–Response Meta-Analysis of Experimental Studies”. In: *Circulation* 143 (16), p. 1542. ISSN: 15244539. DOI: 10.1161/CIRCULATIONAHA.120.050371. URL: <https://pmc.ncbi.nlm.nih.gov/articles/PMC8055199/>.



Gase, Lauren N. et al. (2011). “Estimating the Potential Health Impact and Costs of Implementing a Local Policy for Food Procurement to Reduce the Consumption of Sodium in the County of Los Angeles”. In: *American Journal of Public Health* 101.8, pp. 1501–1507. DOI: 10.2105/AJPH.2011.300138. URL: <https://ajph.aphapublications.org/doi/full/10.2105/AJPH.2011.300138>.



Gaziano, Thomas A. et al. (2025). “Evaluation of South Africa’s Salt Reduction Legislation on Sodium Intake in a Rural Population”. In: *JAMA Cardiology*. URL: <https://jamanetwork.com/journals/jamacardiology/fullarticle/2829791>.



He, Feng J., Sonia Pombo-Rodrigues, and Graham A. MacGregor (2013). “Salt Reduction in England from 2003 to 2011: Its Relationship to Blood Pressure, Stroke and Ischaemic Heart Disease Mortality”. In: *Journal of Human Hypertension*. DOI: 10.1038/jhh.2013.105. URL: <https://www.nature.com/articles/jhh2013105>.



Huang, Liping et al. (2026). “The Effects on Global Health Outcomes of Switching from Regular Salt to Potassium-Enriched Salt: A Modelling Study”. In: *medRxiv*. Preprint. DOI: 10.64898/2026.04.06.26350270. URL: <https://www.medrxiv.org/content/10.64898/2026.04.06.26350270v1>.



Muciño-Sandoval, Karina, Tonatíuh Barrientos-Gutiérrez, et al. (2025). “Modeling the Impact of Implementing PAHO Sodium Reduction Targets in Mexico”. In: *Hypertension*. DOI: 10.1161/HYPERTENSIONAHA.124.24021. URL: <https://www.ahajournals.org/doi/10.1161/HYPERTENSIONAHA.124.24021>.



Neal, Bruce et al. (Sept. 2021). “Effect of Salt Substitution on Cardiovascular Events and Death”. In: *New England Journal of Medicine* 385 (12), pp. 1067–1077. ISSN: 0028-4793. DOI: 10.1056/NEJMOA2105675;ISSUE:ISSUE:DOI. URL: <https://www.nejm.org/doi/pdf/10.1056/NEJMoa2105675>.



Roche, Megan et al. (2026). “Estimated Health and Economic Impact of Taxation on Unhealthy Packaged Foods in India: A Modelling Study”. In: *PLoS Medicine*. DOI: 10.1371/journal.pmed.1004572. URL: <https://journals.plos.org/plosmedicine/article?id=10.1371/journal.pmed.1004572>.



Saxena, Akshar et al. (2025). “Estimated Health Impact, Cost, and Cost-Effectiveness of Taxation on Unhealthy Packaged Foods in the Philippines: A Modelling Study”. In: *The Lancet Public Health*. DOI: 10.1016/S2468-2667(25)00191-4. URL: [https://www.thelancet.com/journals/lanpub/article/PIIS2468-2667\(25\)00191-4/fulltext](https://www.thelancet.com/journals/lanpub/article/PIIS2468-2667(25)00191-4/fulltext).



Song, Jing et al. (2023). “Salt Intake, Blood Pressure and Cardiovascular Disease Mortality in England, 2003–2018”. In: *Journal of Hypertension* 41.11. URL: https://journals.lww.com/jhypertension/abstract/2023/11000/salt_intake,_blood_pressure_and_cardiovascular.6.aspx.



Strauss-Kruger, Michel et al. (2022). “Early Evidence for the Effectiveness of South Africa’s Legislation on Salt Restriction in Foods: The African-PREDICT Study”. In: *Journal of Human Hypertension* 36.6, pp. 545–556. DOI: 10.1038/s41371-021-00653-x. URL: <https://www.nature.com/articles/s41371-021-00653-x>.



Taillie, Lindsey Smith et al. (2024). “Decreases in Purchases of Energy, Sodium, Sugar, and Saturated Fat 3 Years After Implementation of the Chilean Food Labeling and Marketing Law: An Interrupted Time Series Analysis”. In: *PLoS Medicine* 21.1, e1004463. DOI: [10.1371/journal.pmed.1004463](https://doi.org/10.1371/journal.pmed.1004463). URL: <https://journals.plos.org/plosmedicine/article?id=10.1371/journal.pmed.1004463>.